CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

CSUE301 – : Big Data Analytics

Practical File – ODD Semester 2026-27

StudentID: 24DIT021

Student Name: HIT GOAYNI

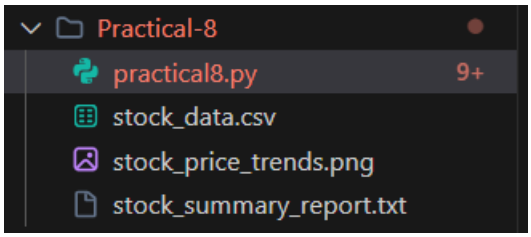
Practical 8

1. Problem Definition

FinEdge Securities Ltd. receives a continuous stream of stock-price records from multiple exchanges. As the volume of market observations grows, conventional processing approaches can become slower for historical and time-based analysis. The objective of this practical is to use PySpark DataFrame aggregations and Window-based operations to summarize stock prices, measure returns and volatility, observe trends, and prepare an educational market analysis report.

PART A — Practical Execution

Step 1: Creating the Practical Workspace and Files



Step 2: Initializing the Spark Session

```
============================================================
 TASK 1: CREATING SPARK SESSION
============================================================
WARNING: Using incubator modules: jdk.incubator.vector
26/09/27 16:41:24 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see
 https://cwiki.apache.org/confluence/display/HADOOP2/WindowsProblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/09/27 16:41:24 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark Version   : 4.2.0
Spark App Name  : Stock-Market-Analytics
Master URL      : local[*]
```

Step 3: Loading the Stock Price Dataset

Step 4: Inspecting the Dataset Schema and Records

Step 5: Grouping Observations by Stock Symbol

Step 6: Computing the Average Stock Price

Step 7: Finding the Maximum Stock Price

Step 8: Finding the Minimum Stock Price

Step 9: Preparing the Stock Summary Report

```
================================================================
 TASK 8: STOCK SUMMARY REPORT
================================================================
+-----+-------------+-------------+-------------+-------------+
|Stock|Total_Records|Average_Price|Minimum_Price|Maximum_Price|
+-----+-------------+-------------+-------------+-------------+
|AAPL |60           |202.22       |181.07       |230.34       |
|AMZN |60           |190.48       |175.0        |205.17       |
|GOOGL|60           |178.72       |159.69       |191.89       |
|MSFT |60           |420.1        |409.53       |435.31       |
|TSLA |60           |246.48       |192.81       |341.0        |
+-----+-------------+-------------+-------------+-------------+
```

Step 10: Determining Period Performance

Step 11: Computing Daily Stock Returns with a Window

```
======================================================================
 SUPPLEMENTARY 1: DAILY STOCK RETURNS
======================================================================
Using Window: Window.partitionBy('Stock').orderBy('Timestamp') with lag('Price')

Sample Daily Returns (First 10 records):
+----------+-----+------+--------------+-----------------------+
|Timestamp |Stock|Price |Previous_Price|Daily_Return_Percentage|
+----------+-----+------+--------------+-----------------------+
|2026-01-05|AAPL |185.0 |NULL          |0.0                    |
|2026-01-06|AAPL |185.03|185.0         |0.02                   |
|2026-01-07|AAPL |182.09|185.03        |-1.59                  |
|2026-01-08|AAPL |182.95|182.09        |0.47                   |
|2026-01-09|AAPL |181.07|182.95        |-1.03                  |
|2026-01-10|AAPL |182.59|181.07        |0.84                   |
|2026-01-11|AAPL |182.41|182.59        |-0.1                   |
|2026-01-12|AAPL |184.01|182.41        |0.88                   |
|2026-01-13|AAPL |186.29|184.01        |1.24                   |
|2026-01-14|AAPL |186.45|186.29        |0.09                   |
+----------+-----+------+--------------+-----------------------+
only showing top 10 rows
```

Step 12: Ranking Stocks by Period Return

```
======================================================================
 SUPPLEMENTARY 2: TOP-PERFORMING STOCK
======================================================================
Ranked Stocks by Cumulative Period Return:
+-----+-----------+----------+-----------------+
|Stock|First_Price|Last_Price|Return_Percentage|
+-----+-----------+----------+-----------------+
|TSLA |210.0      |331.7     |57.95            |
|AAPL |185.0      |230.34    |24.51            |
|GOOGL|160.0      |186.04    |16.27            |
|AMZN |175.0      |193.18    |10.39            |
|MSFT |415.0      |410.46    |-1.09            |
+-----+-----------+----------+-----------------+

Top-Ranked Stock overall : TSLA (57.95%)
```

Step 13: Computing a 3-Period Moving Average

Step 14: Measuring Stock Volatility

```
================================================================
 SUPPLEMENTARY 4: STOCK VOLATILITY
================================================================
Calculating Volatility as Standard Deviation of Daily Returns...
+-----+-----------------+
|Stock|Volatility_StdDev|
+-----+-----------------+
|TSLA |2.46             |
|AMZN |1.38             |
|GOOGL|1.11             |
|AAPL |0.98             |
|MSFT |0.8              |
+-----+-----------------+
```

Step 15: Interpreting the Observed Market Trend

Step 16: Educational Investment Analysis

```
============================================================
 TASK 11: INVESTMENT RECOMMENDATIONS
============================================================
[DISCLAIMER]: Educational analysis only -- not financial advice.

[*] TSLA : Return =  57.95% | Volatility = 2.46% -> Profile: Aggressive Growth (High Return, Elevated Volatility)
[*] AAPL : Return =  24.51% | Volatility = 0.98% -> Profile: Balanced / Moderate (Steady Positive Return, Low Volatility)
[*] GOOGL: Return =  16.27% | Volatility = 1.11% -> Profile: Balanced / Moderate (Steady Positive Return, Low Volatility)
[*] AMZN : Return =  10.39% | Volatility = 1.38% -> Profile: Balanced / Moderate (Steady Positive Return, Low Volatility)
[*] MSFT : Return =  -1.09% | Volatility = 0.80% -> Profile: Defensive / Review (Subdued Return or Higher Fluctuations)
```

Step 17: Visualizing Stock Price Movement

Step 18: Creating the Final Stock Analysis Report

3. Key Questions / Analysis

Q1. Why is Apache Spark useful for financial data analysis?

Apache Spark is useful for financial analytics because it can distribute the processing of large datasets across multiple tasks. DataFrames and built-in functions support efficient filtering, aggregation and time-series calculations, which are common requirements when analysing market data.

Q2. What is the purpose of grouping stock records by symbol?

Grouping records by the Stock column separates the observations belonging to each company. This makes it possible to calculate independent statistics such as average, minimum and maximum price for every symbol.

Spark aggregation functions can then summarize the historical observations for each stock.

For example:

df.groupBy("Stock").agg(avg("Price
")) produces the average price for
every stock symbol.

Q3. How can average stock price be used in analysis?

The average price gives an overall measure of the typical price level during the selected period. It is more informative when considered together with returns, price range and volatility rather than being used as a standalone decision measure.

Q4. What information do maximum and minimum prices provide?

The maximum and minimum values identify the highest and lowest observed prices within the analysed period. Their range helps describe historical price movement and provides context for comparing stocks.

The values should be interpreted together with the time period and other performance measures.

Q5. How can Spark be used for ongoing stock-market monitoring?

Spark can process both historical and streaming data using distributed computation. A Spark-based pipeline can update calculations such as prices, returns, rolling averages and trends as new market records arrive.

4. Conclusion

The practical demonstrated stock-market analytics with PySpark using DataFrame aggregation and Window-based analysis.

The sample dataset contained 300 records for five stock symbols. The data was loaded into a Spark DataFrame, its schema and records were inspected, and average, minimum and maximum prices were calculated for each stock.

Window functions were used to calculate daily returns, period returns and a 3-period moving average. Daily-return standard deviation was also used to compare volatility, while the generated report summarized the observed price behaviour.

A Matplotlib chart was created to visualize stock-price movement over time, and the final analysis was stored in the stock summary report.

Note: The dataset is a generated/sample dataset intended for academic demonstration. The calculated performance profiles are historical observations from the sample and should not be treated as financial advice.